Practical 11: Routing

11(a): Static Routing Configuration

⇒ Aim:

To simulate static Routing Protocol configuration using Cisco packet tracer and verify path selection and failover routes

⇒ Network setup:

- Topology: A network consisting of 3 Routers (0, 1, 2) connected to switches, PCs and a server
- IP addressing scheme:
  - → Router 0 connected to: 10.0.0.0/8, 20.0.0.0/8, 40.0.0.0/8
  - → Router 1 connected to: 20.0.0.0/8, 30.0.0.0/8, 50.0.0.0/8
  - → Router connected to: 40.0.0.0/8, 50.0.0.0/8
- Administrative Distance (AD) setup:
  Primary routes configured with AD 10 and backup routes with AD 20

⇒ Code:

- Router 0 configuration.

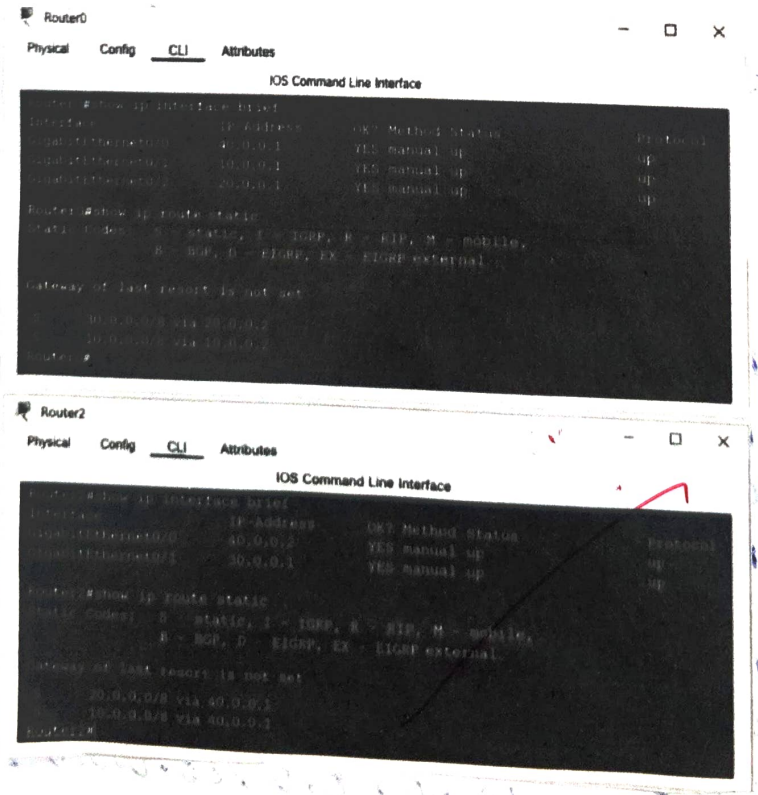
```
Router>enable
Router#configure terminal
Router(config)#ip route 50.0.0.0 255.0.0.0 20.0.0.2 10
Router(config)#ip route 30.0.0.100 255.0.0.0 40.0.0.2 20
Router(config)#ip route 30.0.0.100 255.255.255 40.0.0.2 10
Router(config)#ip route 30.0.0.100 255.255.255 20.0.0.2 20
Router(config)#ip route 50.0.0.0 255.0.0.0 40.0.0.2 10
Router(config)#ip route 50.0.0 255.0.0.0 20.0.0.2 20
Router(config)#ip route#exit
Router#show ip route static
```

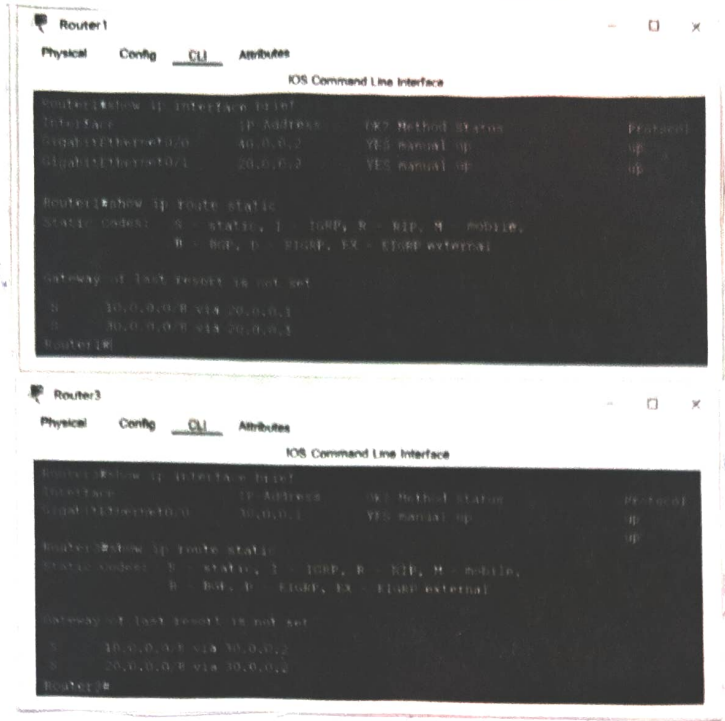
Router 1 Configuration:

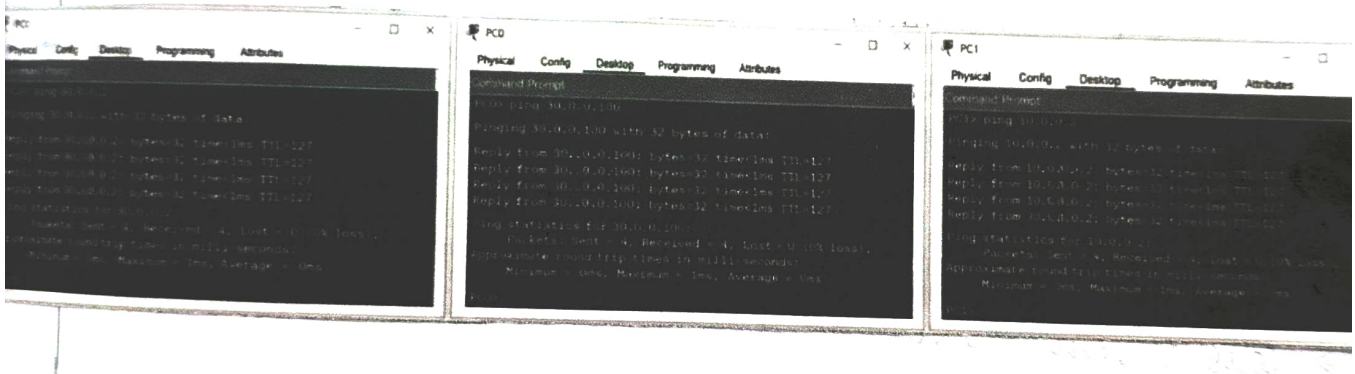
```
Router>enable
Router#config terminal
Router(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1 40
Router(config)#ip route 10.0.0.0 255.0.0.0 50.0.0.1 20
Router(config)#ip route 40.0.0.0 255.0.0.0 20.0.0.1 10
Router(config)#ip route 40.0.0.0 255.0.0.0 50.0.0.1 20
#exit
route static.
```

Router 2 Configuration

```
Router>enable
Router#config Terminal
Router(config)#ip route 10.0.0.0 255.0.0.0 40.0.0.1
Router(config)#ip route 30.0.0.0 255.0.0.0 50.0.0.2
Router(config)#exit
Router#show ip route static
```







Result:

Static routes were successfully configured across all three routers.

11(b): Dynamic Routing using RIP

→ Aim:
To simulate Routing Information Protocol (RIP) using CISCO Packet Tracer and observe dynamic route selection and automatic failover

→ Network Setup:

- Device Interface Assignments

PC0 : 10.0.0.2/8

Pc1 : 20.0.0.2/30

Router 0 serial link:
192.168.1.249 /30

Router1 :
192.168.1.250/30

Router 2:
192.168.1.245/30

→ Code : CISCO IOS commands:

- Router 0:

```
Router >enable
Router #configure terminal
Router (config) #interface fastethernet 0/1
Router (config) #ip address 10.0.0.1 255.0.0.0
Router (config-if) #no shutdown
Router (config-if) #exit
```

• RIP Protocol commands:

!Router 0

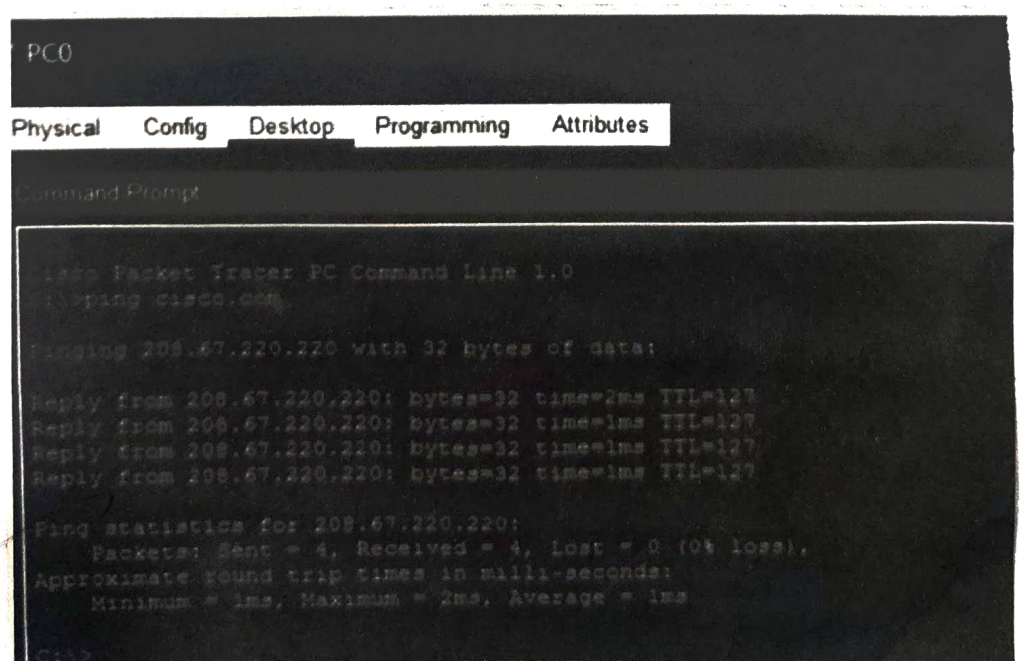
Router0(config)#router rip
Router0(config-router)#network 10.0.0.0
Router(config-router)#network 192.168.1.252
Router(config-router)#network 192.18.1248

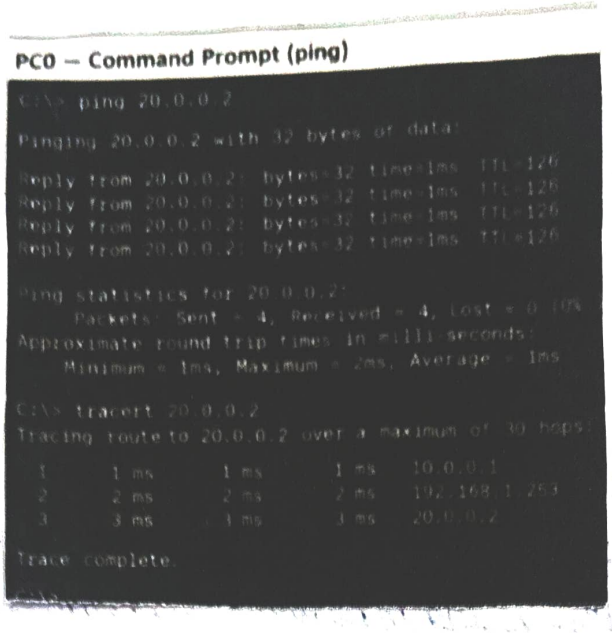
!Router 1

Router1(config)#router rip
Router1(config-router)#network 192.168.1.244
Router1(config-router)#network 192.168.1.248

!Router 2

Router2(config)#router rip
Router2(config)#network 20.0.0.0
Router2(config-router)#network 192.168.1.252
Router2(config-router)#network 192.168.1.244





Result:

Dynamic Routing via RIP was successfully established across all routers.